ENSO Event Forecasting Using Machine Learning with Explainable AI Techniques

I. Reproducibility Instructions

A. Steps to Run

1) Download and place the ENSO (ONI) dataset and ERA5 dataset in the working directory.
2) Open the project in a Python environment (Jupyter Notebook / VS Code / PyCharm).
3) Install the required libraries:

```
pip install numpy pandas xarray scikit-learn matplotlib seaborn shap
```

4) Load the datasets using Python scripts or notebook.
5) Run the preprocessing steps:
   - Convert date format
   - Sort data chronologically
   - Handle missing values
6) Execute feature engineering to create lag features (ONI_lag1, ONI_lag3, ONI_lag6).
7) Train the Random Forest model on the training dataset.
8) Run the forecasting module to generate future ONI predictions.
9) Evaluate the model using RMSE, MAE, and R^2.
10) Visualize the results and generate SHAP plots for interpretation.

B. Libraries

Python, NumPy, Pandas, Scikit-learn, SHAP, Matplotlib

C. Environment

Windows/Linux, 8GB RAM, Python 3.8+